A few farther-sighted trees can improve shallow forests

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Abstract

Looking several splits ahead can improve a decision tree, but the search quickly becomes expensive. Does every tree in a forest need this extra computation? We study mixed forests in which a minority of members optimize their splits over a longer horizon. Repeated paired experiments on PMLB classification tasks compare unpruned trees and forests with matching depth limits and common hyperparameters. Replacing a small fraction of greedy members improves mean accuracy in shallow forests under both feature policies studied. Some mixtures also achieve higher mean accuracy with less fitting work than much larger greedy forests. These gains diminish as the permitted final depth increases, while uniformly farther-sighted induction remains costly. In a separate experiment, training-only tuning of both conventional and mixed-capable families gives the expanded family a higher benchmark-mean accuracy, at substantially greater selection and refit costs. The findings identify selective use of farther-sighted members as a useful option for shallow ensembles and show why split-search depth, tree capacity, and model-selection cost should be evaluated together.

Keywords: Decision trees, Bootstrap ensembles, Bounded split optimization, Computational cost, Statistical evaluation

1. Introduction

Decision trees make a sequence of decisions, yet CART chooses each split by its immediate reduction in impurity [1]. A split that looks unhelpful on its own may reveal a useful interaction after another split. Searching several generations ahead can find such opportunities, but doing so at every node of every tree is expensive. For a forest, this raises a different question: *can a few farther-sighted trees improve the ensemble without paying for farther sight throughout?*

We call a tree k-sighted when it scores a candidate split through the best local subtree of up to k split generations, including the current one. A mixed forest combines members with different horizons. We evaluate this well-studied split-search principle as a way to allocate computation within an ensemble. Two comparisons are kept separate: fixed architectures isolate the effect of replacing greedy members, while training-only tuning compares conventional and mixed-capable model families fairly.

Shallow forests are a natural starting point. Limiting member size can regularize an ensemble, and smaller trees can be beneficial in noisy settings [2, 3]. A depth-three member has at most eight leaves, but a forest of such members can define many more prediction regions. We therefore study shallow ensembles alongside depth-six and unrestricted controls, keeping total depth and the other common settings matched between greedy and farther-sighted learners.

The experiments reveal three useful patterns. First, replacing a minority of greedy members improves shallow-forest mean accuracy under both feature policies studied. Second, a small mixed forest can outperform a much larger greedy forest while using less measured fitting work. Third, allowing farther-sighted candidates in training-only model selection increases benchmark-mean accuracy, although the search for those candidates is much more costly. Increasing every member's horizon is therefore a different proposition from using farther sight selectively. The depth comparisons further show that the benefits are strongest when tree capacity is constrained.

The study uses repeated paired partitions of 57 PMLB classification tasks and reports predictive differences, uncertainty, fitting work, and selection cost. Code and a guide to reproducing the results are available at https://github.com/MathiasValla/non-greedy-decision-trees.

2. Related work

Greedy induction and bootstrap aggregation provide the foundations for the models studied here [1, 4, 5]. Random forests additionally randomize the features available at each split [6].

Non-greedy induction predates the present study. Norton's overlapping search commits the shallowest choice and searches again below it [7], while Ragavan and Rendell [8] used lookahead for feature construction. Murthy and Salzberg's LOOK procedure scores a split with optimally split children, corresponding to the two-generation case here [9]. Their experiments already question whether added computation reliably improves prediction. Later work revisited pathology [10]. Esmeir and Markovitch [11] studied anytime induction, including bagged non-greedy learners and the tradeoff between computation per tree and committee size. Norouzi et al. [12] jointly

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optimize oblique split and leaf parameters, a different form of non-greediness.

A direct forest precedent is the stepwise lookahead forest of Donick and Lera [13], which optimizes two-level blocks to recover interactions in noisy data. Our builder commits one root split and repeats the search at each child; the ensemble question is how to distribute horizons across members. Randomized splits [14] and random final depths [15] offer other ways to vary members. The relationship between ensemble diversity and accuracy has itself been studied extensively [16]. Here we measure the effect of changing horizon allocation directly.

Modern optimal-tree methods exploit mathematical programming [17, 18, 24], bounds and sparse objectives [19, 20], and cached or separable subproblems in DL8.5, MurTree, and STreeD [21, 22, 23]. Reference ensembles can guide restricted search [25]. SPLIT and LicketySPLIT combine non-greedy choices with greedy continuation [26], using objectives and continuation rules different from the truncated terminal impurity considered here. Recent comparisons further show how objectives, size constraints, and tuning affect the relative performance of greedy and optimal trees [27]. These developments place our contribution in empirical evaluation of horizon allocation, rather than in proposing another optimal-tree solver.

2.1. Tree-size regularization

Tree size is also a regularization choice. Zhou and Mentch [2] report benefits from smaller random-forest members at low signal-to-noise ratios. Duroux and Scornet [3] study subsampling and early termination theoretically for quantile-split regression forests and empirically for Breiman-style regression forests. These regression studies vary leaf limits and resampling; Duroux and Scornet's small-tree empirical alternative also removes bootstrap resampling. They motivate studying constrained member size, rather than prescribing a universal classification-tree depth. Competitive small classification trees [17] provide a related motivation. At the other end of the spectrum, consistency results for random forests study trees whose size can grow with the sample [28]. Our comparison of shallow, deeper, and unrestricted bootstrap Gini classifiers examines how the benefit of split optimization changes with capacity.

3. Method

3.1. Horizon and final depth

Let S be the examples reaching a node, $I(S)$ its Gini impurity, and $W(S)$ its total weight. Let D be the maximum number of splits along a final root-to-leaf path and d the current depth. At that node the actual search horizon is $h = \min(k, D - d)$; when D is unrestricted, $h = k$. Increasing k changes search, whereas increasing D changes capacity.

Define the optimal local terminal score recursively:

$$C_0(S) = W(S)I(S), \tag{1}$$

$$C_h(S) = \min\left\{C_0(S),\ \min_{s\in\mathcal{S}(S)}\left[C_{h-1}(S_l(s)) + C_{h-1}(S_r(s))\right]\right\}. \tag{2}$$

Here $\mathcal{S}(S)$ contains admissible axis-aligned midpoint thresholds between distinct observed feature values, subject to the leaf, depth, and feature constraints. The selected root minimizes the second term when it is no worse than stopping, up to numerical tolerance. Feasible zero-gain splits are allowed, as in unpruned CART. Only that root is committed. The configured horizon is then restored at each actual child. This is receding-horizon induction, not a single depth-k fit followed by an unchanged completion.

For a fixed node, expanding the feasible horizon cannot worsen the optimal terminal training impurity. With all features and $k \geq D$ at the root, the search covers the depth-bounded terminal-impurity problem; with feature subsampling it is conditional on the seeded feature policy. This makes $k = D = 3$ an instructive full-horizon case. The experiments ask how these local training improvements translate into prediction and cost.

3.2. Matched trees and mixed forests

The one-sighted baseline is scikit-learn CART. All learners use Gini impurity, the same permitted total depth, split and leaf minima, and feature policy. None is pruned, and feasible zero-gain splits are allowed. Every distinct admissible numerical threshold is considered; there is no candidate cap. At each node we consider either all p features or an initially sampled subset of $\max(1, \lfloor \sqrt{p} \rfloor)$ features, examining more if that subset provides no admissible split.

A mixed forest replaces designated CART members with farther-sighted trees, using the same bootstrap draw and seed for each replacement. Every ensemble position contains exactly one member. Bootstrap multiplicities are sample weights, and the sample-count constraints exclude zero-weight rows. Horizons are assigned without using class labels, in blocks of 20 so that every evaluated ensemble size has the specified composition. Predictions average class-aligned member probabilities. The all-greedy construction was checked against the library random forest over the common parameter grid.

We describe a mixture by its member proportions and, for convenience, its mean horizon $\bar{k} = \sum_j p_j k_j$. The proportions matter: 90% one-sighted plus 10% two-sighted and 95% one-sighted plus 5% three-sighted both have $\bar{k} = 1.1$, but involve different searches.

3.3. Search cost

The two child optima in Eq. (2) can be computed separately. If every recursive call has at most m candidates and E_h counts candidate evaluations, a worst-case bound is

$$E_0 = 0, \qquad E_h \leq m + 2mE_{h-1}, \qquad E_h \leq m\frac{(2m)^h - 1}{2m - 1}. \tag{3}$$

The bound is exponential in local horizon. Final fitting sums such searches over the committed nodes, with actual work also depending on sample sizes and stopping. A fixed short horizon can therefore build a deep tree; increasing the final-depth limit does not itself increase k.

4. Experimental protocol

4.1. Dataset sample and repeated partitions

We use a tractable 57-dataset sample of PMLB classification tasks [29]. All methods share PMLB's supplied numeric representation and categorical encodings. A common complete-case rule removes eleven rows from `penguins`; no imputation or scaling is applied. Dataset names, dimensions, class counts, and preprocessing records are available with the code.

Each dataset receives five paired stratified 75/25 outer train–test partitions with fixed, reproducible seeds. The seeds differ between repetitions, as do estimator and bootstrap seeds. All methods share the same partitions within a repetition.

4.2. Models and training-only selection

Fixed-architecture comparisons use $D \in \{3, 6, \infty\}$, minimum leaf size one, and either all features or square-root feature subsampling. Single trees use $k = 1, 2, 3$ with identical common settings and no pruning. Every forest composition is evaluated at $T \in \{20, 40, 60, 100, 200\}$. In addition to the three pure horizons, each pair $(1, 2)$, $(1, 3)$, and $(2, 3)$ uses 5%, 10%, 25%, or 50% of the farther-sighted horizon. A sparse three-way composition uses $85\%, 10\%, 5\%$ of horizons one, two, and three. Thus sixteen explicitly specified compositions span every ensemble size; equal average horizons do not identify identical mixtures.

In the separate tuned experiment, three model families receive training-only inner cross-validation. The conventional greedy family (RF) contains only CART members. Its feature setting is tuned: a random subset at each split gives a feature-subsampled random forest, while using all features gives bagged CART. The 1/2 family allows pure forests and mixtures of one- and two-sighted members. The inclusive 1/2/3 family allows the full sixteen-composition grid. Thus both expanded families can also select a conventional greedy forest.

All three families share depths $\{3, 6, \infty\}$, leaf minima $\{1, 5\}$, the two feature policies, and ensemble sizes $\{20, 40, 60, 100\}$. They have 48, 288, and 768 candidate configurations, respectively. The 200-member models appear only in the fixed experiment. Training-only selection keeps the outer test data separate from hyperparameter choice [30].

Inner folds are shared across families. Shuffled stratified cross-validation uses $\min(3, c_{\min})$ folds, where $c_{\min}$ is the smallest observed outer-training class count. A singleton or single-class training partition instead uses ordinary shuffled three-fold cross-validation. Mean inner accuracy chooses the model; ties favour lower measured member-fitting work, then a fixed candidate ordering. The chosen configurations are refitted separately on the outer-training data before test evaluation. Three-sighted members are included in both fixed and tuned experiments.

4.3. Timing and outcomes

Accuracy is the primary outcome; balanced accuracy, the mean recall over classes present in the scoring partition, assesses sensitivity to class imbalance. Scores and costs are first averaged over repetitions within each dataset, then equally across datasets.

For fixed forests, fitting work is the sum of measured fitting durations of the included members. Fixed single-tree timings likewise cover fitting alone. Both exclude construction, bootstrap preparation, and prediction. For tuned models, we report the selected model's direct refit wall time, including construction and bootstrap preparation, separately from cross-validation fitting work. The latter includes the tree fits needed to evaluate the whole candidate grid, even when a cheap model is ultimately selected. Reused fits are counted once within each family's search, and each family is charged the work it would require if run on its own. Recorded workflow-stage wall times are also supplied; they exclude checkpoint input/output, startup, downloads, and idle gaps.

Experiments use five concurrent, single-threaded workers on a MacBook Air with an eight-core Apple M3 processor and 16 GB memory. Numerical figures are computed from experimental results using reproducible NumPy, SciPy, pandas, and Matplotlib workflows. OpenAI Codex (OpenAI) assisted with benchmark, analysis, and plotting code. Historical application and model versions were not recorded; the plots are data-derived and use no generative image model.

4.4. Uncertainty and multiple comparisons

For each contrast, paired repetition differences are averaged within datasets. A percentile bootstrap with 20,000 resamples of the dataset means gives a pointwise, unadjusted 95% interval for the mean effect. Two-sided Wilcoxon signed-rank tests compare these same dataset-level differences [31]; our calculation omits zero differences. Mean-effect intervals and signed-rank tests describe different aspects of the effect distribution: its average and its location under the usual symmetry assumption.

One Holm adjustment [32] covers eleven primary contrasts: eight fixed 100-member comparisons of 10% two- or three-sighted replacements with matched greedy forests, at depth three or unrestricted depth and either feature setting; and three pairwise comparisons of the tuned families. We denote the adjusted values by p_H. Other depths, sizes, proportions, and single-tree contrasts are descriptive.

Two sensitivity analyses change benchmark weighting or membership. One gives equal weight to 48 known task families, averaging related GAMETES, MONK, LED, parity, and Cy Young tasks first. The other removes 19 explicitly named synthetic/constructed tasks, leaving 38 tasks. The full membership lists are supplied with the results.

5. Results

Table 1 summarizes representative fixed architectures and the three tuned families. Table 2 reports the eleven primary paired contrasts. Accuracy differences below are percentage points; brackets give the mean-effect intervals defined in Section 4.4.

Table 1: Performance averaged over 57 datasets and five paired repetitions. Fixed rows use all features, minimum leaf size one, and 100 forest members; replacement forests contain 90% CART and 10% of the indicated horizon. Accuracy and balanced accuracy are proportions. Fixed-tree costs measure fitting alone; fixed-forest costs sum member fits. Tuned-family costs measure direct refitting, including construction/bootstrap preparation and excluding prediction and selection. Section 4.3 defines the timing scopes.

Model	D	Accuracy	Balanced accuracy	Fit measure (s)	Realized depth
CART tree	3	0.6782	0.6241	0.0010	3.0
Tree $k = 2$	3	0.7059	0.6543	0.0282	3.0
Tree $k = 3$	3	0.7136	0.6647	1.6845	3.0
Pure $k = 1$ forest	3	0.7288	0.6721	0.0677	2.9
Pure $k = 2$ forest	3	0.7478	0.6931	1.6807	3.0
Pure $k = 3$ forest	3	0.7522	0.6959	80.2307	3.0
10% $k = 2$ replacement	3	0.7393	0.6835	0.2293	2.9
10% $k = 3$ replacement	3	0.7415	0.6853	8.0933	2.9
CART tree	6	0.7282	0.6947	0.0011	5.6
Tree $k = 2$	6	0.7321	0.6977	0.0569	5.9
Tree $k = 3$	6	0.7341	0.7009	3.7724	5.9
Pure $k = 1$ forest	6	0.7665	0.7297	0.0797	5.4
Pure $k = 2$ forest	6	0.7606	0.7201	3.7387	5.9
Pure $k = 3$ forest	6	0.7556	0.7166	200.3026	5.9
10% $k = 2$ replacement	6	0.7692	0.7321	0.4463	5.4
10% $k = 3$ replacement	6	0.7723	0.7350	20.1196	5.4
CART tree	∞	0.7451	0.7194	0.0017	10.1
Tree $k = 2$	∞	0.6938	0.6578	0.1592	15.1
Tree $k = 3$	∞	0.6622	0.6208	7.9657	17.0
Pure $k = 1$ forest	∞	0.7752	0.7450	0.1466	8.9
Pure $k = 2$ forest	∞	0.7069	0.6615	9.7728	13.2
Pure $k = 3$ forest	∞	0.6784	0.6302	387.2398	14.9
10% $k = 2$ replacement	∞	0.7758	0.7442	1.1200	9.4
10% $k = 3$ replacement	∞	0.7750	0.7430	38.5187	9.5
Selected RF family	selected	0.7899	0.7533	0.0626	6.5
Selected 1/2 family	selected	0.7996	0.7642	0.9518	5.5
Selected inclusive family	selected	0.8056	0.7693	12.9374	5.4

5.1. All-feature single trees and pure forests

At $D = 3$, mean accuracies for single trees with $k = 1, 2, 3$ are $67.82\%, 70.59\%, 71.36\%$. The paired horizon-two/three gains over CART are 2.77 points [0.68, 5.22] and 3.54 points [1.45, 6.01], but medians are only 0.13 and 0.40 points, with 29/3/25 and 31/3/23 wins/ties/losses. Mean fit-only times are 0.00103, 0.0282, and 1.684 s, respectively. At $D = 6$, mean differences shrink to 0.39/0.59 points and medians become $-0.48/-1.27$ points. Without a depth limit, mean differences are -5.13 [-8.69, -1.84] and -8.29 [-12.45, -4.39] points. Mean realized depths then increase from 10.1 for CART to 15.1/17.0 for horizons two/three. Figure 1 shows this change from gains at shallow depth to losses without a depth limit.

For pure $T = 100$ forests with all features, horizon-two/three mean differences from the matched CART forest are 1.90/2.34 points at $D = 3$, $-0.58/-1.09$ at $D = 6$, and $-6.84/-9.68$ without a depth limit. Shallow mean fitting work is 0.0677, 1.6807, and 80.2307 s for the three pure horizons. With square-root feature subsampling, the corresponding mean differences are 2.38/2.85, 1.27/1.96, and 1.08/0.78 points. The last two intervals are [0.35, 1.88] and [-0.75, 2.44], with medians 0.29 and zero points. Thus the depth pattern differs between the two feature policies, with every contrast using its own matched CART control.

5.2. Sparse horizon replacements

At $T = 100$, $D = 3$, with all features, replacing ten CART members with horizon-two or horizon-three members gives mean gains of 1.06 points [0.45, 1.77] and 1.27 points [0.61, 2.02]. Both medians are zero; wins/ties/losses are 27/21/9 and 28/14/15. Their joint-Holm signed-rank values are $p_H = 0.00996$ and 0.02066. Balanced-accuracy gains are 1.15/1.33 points. Mean member-fitting work rises from 0.0677 s for CART to 0.2293/8.0933 s, or 3.39/119.52 times the control. The 85/10/5% composition has a mean gain of 1.36 points [0.52, 2.30] at 4.2260 s.

With square-root features, the two shallow replacement gains are 0.42 [0.07, 0.78] and 1.04 [0.46, 1.74] points, with $p_H = 0.09539$ and 0.00150. Their medians are zero and 0.13 points. Without a depth limit, the four primary replacement mean gains are $0.05/-0.02$ points with all features and 0.22/0.28 points with square-root features. Their medians are zero, and adjusted values range from 0.34732 to 1. At the depth-six all-feature setting, the replacement mean differences are 0.27/0.58 points, again with zero medians.

Table 2: The eleven primary paired comparisons. Fixed rows replace ten of 100 CART members with the indicated horizon, at minimum leaf size one and the stated feature setting. Effects and pointwise, unadjusted 95% mean-effect intervals are in percentage points. p_H is the two-sided Wilcoxon signed-rank value Holm-adjusted jointly over these eleven contrasts. W/T/L counts dataset-mean wins/ties/losses for the first model. Cost ratios divide equal-dataset mean costs: summed member-fitting work for the eight fixed rows, direct selected-model refit wall time for the three selected-family rows. Selection work is reported separately.

Comparison	D	Mean difference [95% interval]	p_H	W/T/L	Cost ratio
10% $k = 2$ vs. CART (all)	3	1.06 [0.45, 1.77]	0.00996	27/21/9	3.39
10% $k = 3$ vs. CART (all)	3	1.27 [0.61, 2.02]	0.02066	28/14/15	119.52
10% $k = 2$ vs. CART (sqrt)	3	0.42 [0.07, 0.78]	0.09539	27/22/8	4.84
10% $k = 3$ vs. CART (sqrt)	3	1.04 [0.46, 1.74]	0.00150	32/19/6	106.73
10% $k = 2$ vs. CART (all)	∞	0.05 [−0.27, 0.44]	1.00000	19/20/18	7.64
10% $k = 3$ vs. CART (all)	∞	−0.02 [−0.49, 0.55]	1.00000	16/20/21	262.81
10% $k = 2$ vs. CART (sqrt)	∞	0.22 [−0.02, 0.47]	0.75287	23/18/16	18.90
10% $k = 3$ vs. CART (sqrt)	∞	0.28 [0.00047, 0.59]	0.34732	25/16/16	367.04
Selected 1/2 vs. RF	selected	0.97 [−0.07, 2.30]	1.00000	29/8/20	15.20
Selected inclusive vs. RF	selected	1.57 [0.30, 3.28]	0.48838	28/11/18	206.61
Selected inclusive vs. 1/2	selected	0.60 [−0.11, 1.63]	1.00000	18/23/16	13.59

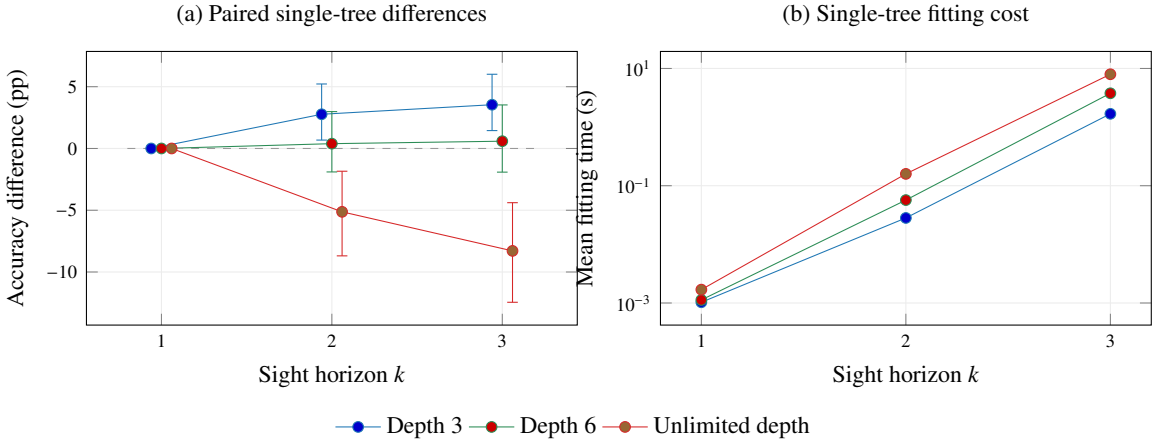


Figure 1: Depth-matched unpruned single trees with all features. Left: mean paired accuracy differences from CART, with pointwise, unadjusted 95% dataset-bootstrap intervals. Right: mean fitting duration, excluding construction and prediction. Sight horizon varies independently of permitted final depth.

For the two shallow all-feature anchors, equal weighting of the 48 known dataset families reduces the means to 0.77/0.91 points, with intervals [0.25, 1.45] and [0.32, 1.63]. On the 38 tasks outside the named synthetic/constructed set, the means are 0.53/0.63 points, with intervals [0.06, 1.22] and [0.07, 1.35].

5.3. *Descriptive mean-cost curves*

Figure 2 shows why mixing is worth considering. A 20-member forest with five two-sighted trees and fifteen CART trees has 74.06% mean accuracy at 0.0955 s of fitting work. The 200-member pure CART forest has 73.02% accuracy at 0.1338 s. In this example, replacing a few members is more effective than adding many greedy trees: mean accuracy is 1.04 points higher with about 29% less fitting work. The 20-member mixture with only two two-sighted members reaches 73.37% mean accuracy at 0.0470 s of fitting work, also exceeding the 200-member greedy forest at lower work.

Mixtures occupy intermediate positions between the cheap greedy curve and the expensive uniformly farther-sighted curves. A 100-member forest with 25% two-sighted members reaches 74.57% at 0.4709 s. Smaller pure farther-sighted forests are competitive too: below the 0.7 s mean-work reference, the highest observed mean is 74.79% for 20 two-sighted members at 0.3381 s. Every plotted composition containing three-sighted members lies above that reference. The useful choice is therefore between ensemble size and composition, rather than a uniform increase in either tree count or horizon.

5.4. *Training-selected families*

Mean accuracy is 78.99% for the RF family, 79.96% for the one-/two-sighted family, and 80.56% for the inclusive family (Figure 3). Relative to tuned RF, the 1/2 family gains 0.97 points [−0.07, 2.30]. The inclusive family gains 1.57 points [0.30, 3.28], and its balanced-accuracy gain is 1.60 points. Inclusive versus 1/2 gives 0.60 points [−0.11, 1.63]. The tuned contrasts' wins/ties/losses and adjusted tests are reported in Table 2.

Selection does not always choose a mixed model. Among the 285 inclusive choices, 133 are pure forests (61/28/44 at horizons one/two/three) and 152 are mixed (146 use two horizons and six use all three). The one-/two-sighted family selects 168 pure forests and 117 mixtures. Depth-three/six/unrestricted

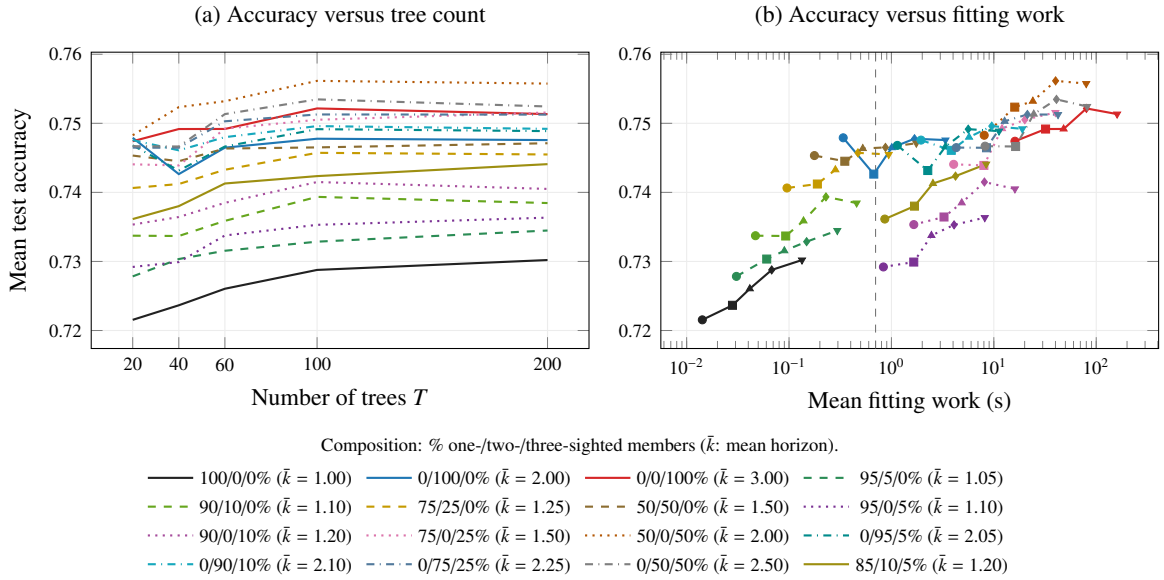


Figure 2: Accuracy–fitting-work curves for depth-three, all-feature forests on 57 datasets and five paired repetitions. Every composition spans 20, 40, 60, 100, and 200 members, marked respectively by circles, squares, upward triangles, diamonds, and downward triangles in the right panel. The dashed line marks 0.7 s of benchmark-mean member-fitting work. Proportions identify each mixture; $\bar{k}$ is its mean horizon. These descriptive curves report fixed architectures, excluding selection cost.

selection counts are 66/87/132 for RF, 90/106/89 for the one-/two-sighted family, and 104/103/78 for the inclusive family. Both feature policies are selected in every family.

5.5. The cost of model selection

Standalone cross-validation fitting work averages 2.271, 112.079, and 2,671.049 s for RF, one-/two-sighted, and inclusive families. Direct refit wall times, including construction/bootstrap preparation but not test prediction, are 0.0626, 0.9518, and 12.9374 s. Their ratios to RF are 15.20 and 206.61 for the two expanded families; these are refit ratios, not selection-cost ratios. Mean sums of measured workflow stage wall times are 3.405, 118.035, and 2,693.803 s. Selection requires much more work than fitting the eventual model, especially when three-sighted candidates are included.

6. Discussion

The main practical finding is that farther sight need not be an all-or-nothing choice. Sparse replacements improve shallow-forest mean accuracy under both feature policies, and three of the four primary shallow contrasts retain $p_H < 0.05$. Figure 2 shows the complementary computation argument: adding a few two-sighted members can achieve more than adding many greedy members. The benefit of a mixed forest is the option to spend more search effort on only part of the ensemble.

The tuned comparison strengthens the predictive finding under the original benchmark weighting. The inclusive family's mean-effect interval lies above zero even though the conventional baseline is tuned on the same common parameter grid. Its higher mean accuracy is therefore not simply an artefact of comparing tuned mixtures with an untuned RF. Selection often chooses an actual mixture, but also chooses pure forests. The tuned result concerns the value of the expanded family; the fixed comparisons isolate the effect of replacing specified greedy members with farther-sighted members.

The depth controls indicate where the observed benefits are strongest. When each member has few splits available, selecting those splits more carefully can help. With more splits available, greedy induction may recover useful partitions later in the tree. Fixed greedy forests improve at deeper capacities, and tuned RF selects depth six or unrestricted depth in 219 of 285 outer dataset–repetition selections. The unrestricted all-feature farther-sighted trees grow deeper yet predict less accurately, a pattern consistent with overfitting. Optimizing local terminal impurity does not penalize tree size; a more thorough search can therefore fit sample-specific structure. This is a plausible explanation to test with training–validation curves and explicit regularization. Feature-subsampled forests retain small positive mean gains at unrestricted depth, so the feature policy also matters.

6.1. Scope and limitations

The gains are heterogeneous. For tuned inclusive versus RF, the mean gain is positive but the median is zero and the adjusted signed-rank test does not reject its location null. Also, we note that a nonsignificant test does not show that two methods perform equally. The inclusive mean gain becomes 1.03 points [−0.27, 2.90] under equal weighting of 48 known task families and 0.53 [−0.91, 2.77] on the 38 tasks outside the named synthetic/constructed set. The corresponding 1/2 effects are 0.61 [−0.32, 2.02] and 0.38 [−0.66, 2.09]. These sensitivities show how the benchmark's composition affects the estimated

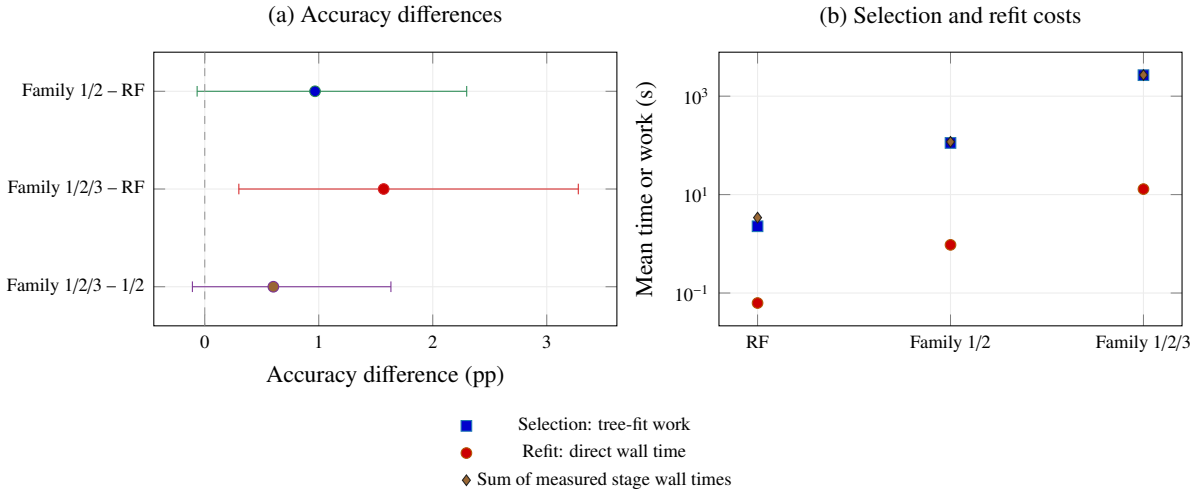


Figure 3: Tuned forest families on 57 datasets and five paired repetitions. Family 1/2 allows horizons one and two; the inclusive 1/2/3 family allows three as well. Both can select pure forests. Left: mean paired accuracy differences with pointwise, unadjusted 95% dataset-bootstrap intervals. Right: cross-validation tree-fitting work, direct selected-model refit wall time, and summed workflow-stage wall times, with the scopes defined in Section 4.3.

average benefit. The sample is rather small and includes related and constructed tasks; overlapping holdouts are averaged within tasks, not treated as additional independent datasets.

Computation is the other important limit. Three-sighted search and full selection over expanded families are expensive. The common hyperparameter grid ensures matched choices, but the mixed families have more candidates to evaluate. Figure 2 compares benchmark means: the illustrated mixtures beat particular greedy forests at lower measured fitting work, not necessarily the best pure forest at every budget. Connecting plotted points does not evaluate an intermediate model, and the curve excludes selection work. A per-dataset fixed-budget policy would need training-only selection and a check that the final fit respects the budget. The timings also reflect this implementation and concurrent hardware load, not an intrinsic cost of every non-greedy method. Larger independent tasks and comparisons with efficient bounded or hybrid search [21, 22, 23, 26] would establish how far the observed tradeoffs extend.

7. Conclusion

Across the tested depth-three settings, sparse replacements consistently yielded higher observed benchmark-mean accuracy than their matched greedy controls. They also offer a useful accuracy–fitting-work compromise: a small mixture can outperform a much larger greedy forest while requiring less member-fitting work. Allowing farther-sighted candidates in training-only tuning with matched common hyperparameters raises benchmark-mean accuracy as well, although selection is substantially more expensive. Together, these results support allocating farther sight selectively rather than assuming that every member should search farther. Larger independent, resource-constrained comparisons with modern search methods are needed for a more general conclusion. This study provides a concrete empirical step: mixed forests can improve on larger pure greedy forests at lower benchmark-mean fitting work.

CRediT authorship contribution statement

Mathias Valla: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing–original draft, Writing–review and editing.

Declaration of competing interest

The author declares no competing interests.

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Data and code availability

The datasets are publicly available from PMLB [29]. Code, repeat-level results, selected hyperparameters, paired analyses, figures, and manuscript sources are available at https://github.com/MathiasValla/non-greedy-decision-trees. The repository's `paper/REPRODUCING_RESULTS.md` documents how to retrieve and reproduce every result in this article.

Declaration of generative AI and AI-assisted technologies

OpenAI Codex (OpenAI) assisted with literature discovery, manuscript revision, and development of benchmark, analysis, and plotting code. Figures are computed from experimental results without generative image tools. Responsibility for the submitted article rests with the author.

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